

Notice SAE

Préparation

- Se connecter à un réseau local
- Prendre connaissance de votre adresse IP :

```
ip a
```

- Noter le masque de sous-réseau

Découvrir les machines du réseau

- Ouvrir un terminal
- Installer **nmap** si ce n'est pas déjà fait :

```
sudo apt install nmap
```

- Lancer une analyse du réseau :

```
sudo nmap -sn <votre_adresse_ip>/<masque_de_sous_reseau>
```

Exemple :

```
adminetu@RT007:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eno1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 84:69:93:07:a1:5c brd ff:ff:ff:ff:ff:ff
    altname enp0s31f6
    inet 192.168.111.117/24 brd 192.168.111.255 scope global dynamic noprefixroute eno1
        valid_lft 82034sec preferred_lft 82034sec
    inet6 fe80::8a8e:a6c7:da5d:4282/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 26:9e:8e:6e:24:2b brd ff:ff:ff:ff:ff:ff
```

```
adminetu@RT007:~$ sudo nmap -sn 192.168.111.117/24
[sudo] Mot de passe de adminetu :
Starting Nmap 7.93 ( https://nmap.org ) at 2025-10-17 10:32 CEST
Nmap scan report for 192.168.111.1
Host is up (0.00044s latency).
MAC Address: 84:A0:6E:98:5B:BD (Sagemcom Broadband SAS)
Nmap scan report for RTP21 (192.168.111.68)
Host is up (0.00033s latency).
MAC Address: 84:69:93:0A:01:C4 (HP)
Nmap scan report for 192.168.111.71
Host is up (0.00020s latency).
MAC Address: 84:69:93:0A:01:AC (HP)
Nmap scan report for RTP35 (192.168.111.72)
Host is up (0.00030s latency).
MAC Address: 84:69:93:07:A3:51 (HP)
Nmap scan report for 192.168.111.73
Host is up (0.00026s latency).
MAC Address: 84:69:93:07:A4:57 (HP)
Nmap scan report for RTP34 (192.168.111.77)
Host is up (0.00031s latency).
MAC Address: 84:69:93:07:A3:6B (HP)
Nmap scan report for RTP23 (192.168.111.80)
Host is up (0.00035s latency).
MAC Address: 84:69:93:06:51:F1 (HP)
```

Découvrir les OS des machines du réseau (Fingerprint)

- Lancer une analyse du réseau (plus avancée):

```
sudo nmap -O <votre_adresse_ip>/<masque_de_sous_réseau>
```

Exemple :

```
adminetu@RT007:~$ sudo nmap -O 192.168.111.117/24
Starting Nmap 7.93 ( https://nmap.org ) at 2025-10-17 10:40 CEST
Nmap scan report for 192.168.111.1
Host is up (0.00085s latency).
Not shown: 845 closed tcp ports (reset), 148 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
139/tcp   open  netbios-ssn
443/tcp   open  https
445/tcp   open  microsoft-ds
5060/tcp  open  sip
9000/tcp  open  cslistener
MAC Address: 84:A0:6E:98:5B:BD (Sagemcom Broadband SAS)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.93%E=4%D=10/17%OT=53%CT=135%CU=42097%PV=Y%DS=1%DC=D%G=Y%M=84A06
OS:E%TM=68F20121%P=x86_64-pc-linux-gnu)SEQ(SP=C8%GCD=1%ISR=CE%TI=Z%CI=Z%II=
OS:I%TS=U)OPS(O1=M5B4NNSNW4%O2=M5B4NNSNW4%O3=M5B4NW4%O4=M5B4NNSNW4%O5=M5B4N
OS:NSNW4%O6=M5B4NNS)WIN(W1=16D0%W2=16D0%W3=16D0%W4=16D0%W5=16D0%W6=16D0)ECN
OS:(R=Y%DF=Y%T=40%W=16D0%O=M5B4NNSNW4%CC=N%Q=)T1(R=Y%DF=Y%T=40%S=O%A=S+%F=A
OS:S%RD=0%Q=)T2(R=N)T3(R=Y%DF=Y%T=40%W=16D0%S=O%A=S+%F=AS%O=M5B4NNSNW4%RD=0
OS:%Q=)T4(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z
OS:%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=N
OS: )U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%
OS:DFI=N%T=40%CD=S)

Network Distance: 1 hop

Nmap scan report for RTP21 (192.168.111.68)
Host is up (0.00057s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
902/tcp   open  iss-realsecure
MAC Address: 84:69:93:0A:01:C4 (HP)
Device type: general purpose
Running: Linux 5.X
OS CPE: cpe:/o:linux:linux_kernel:5
OS details: Linux 5.0 - 5.3
```

Découvrir les services disponibles sur une machine distante

- Lancer une analyse des ports et services ouverts sur une machine distante :

```
sudo nmap -sV <adresse_ip_cible>
```

Exemple :

```
adminetu@RT007:~$ sudo nmap -sV 192.168.111.127
Starting Nmap 7.93 ( https://nmap.org ) at 2025-10-17 10:46 CEST
Nmap scan report for 192.168.111.127
Host is up (0.00017s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 9.2p1 Debian 2+deb12u7 (protocol 2.0)
902/tcp   open  ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
MAC Address: 84:69:93:07:A4:8A (HP)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.63 seconds
```

Découvrir les versions des services disponibles et expliquer le fonctionnement de la recherche

- Lancer une analyse des ports et services ouverts sur une machine distante avec en plus la détection de version :

```
sudo nmap -sV --version-all <adresse_ip_cible>
```

Exemple :

```
adminetu@RT007:~$ sudo nmap -sV --version-all 192.168.111.127
Starting Nmap 7.93 ( https://nmap.org ) at 2025-10-17 10:50 CEST
Nmap scan report for 192.168.111.127
Host is up (0.00042s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 9.2p1 Debian 2+deb12u7 (protocol 2.0)
902/tcp   open  ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
MAC Address: 84:69:93:07:A4:8A (HP)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.57 seconds
```